

A Curvature Decomposition of the Explicit Formula for the Riemann Zeta Function

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Abstract

We study the second logarithmic derivative of the completed Riemann zeta function in the form

$$H_\xi(\sigma, t) := \partial_\sigma^2 \log |\xi(\sigma + it)|^2,$$

and interpret it as a second-derivative specialisation of the classical explicit formula. This yields a prime-cutoff decomposition $H_\xi = H_{\text{local}} + H_{\text{dual}}$ into local curvature contributions from the archimedean place and the finite Euler factors, together with a bookkeeping remainder H_{dual} defined as the finite-cutoff difference $H_\xi - H_{\text{local}}$. The note also discusses the automorphic Rankin–Selberg analogue and a structural comparison of the curvature field with Li-type positivity criteria, noting that local positivity does not imply global Li positivity.

What is proved. Local positivity at every finite place: $V_p(\sigma) \geq 0$ for all primes p and all $\sigma > 0$ (Observation 3.1). Together with the classical archimedean log-convexity $\psi_1(\sigma) \geq 0$, this yields non-negativity of the truncated local curvature $H_{\text{local}}(\sigma, \kappa) \geq 0$ (Corollary 3.2). Critical divergence on the line $\sigma = \frac{1}{2}$: $H_{\text{local}}(\frac{1}{2}, \kappa) \asymp (\log \kappa)^2$ as $\kappa \rightarrow \infty$ (Lemma 4.1). Subcritical convergence: $H_{\text{local}}(\sigma, \kappa)$ tends to a finite limit as $\kappa \rightarrow \infty$ for every $\sigma > \frac{1}{2}$ (Lemma 4.2).

What is numerical. At a representative off-critical point $(\sigma, t) = (0.3, 0)$ with $\kappa = 1300$, $H_{\text{local}} \approx 915$ while $H_\xi \approx 0.09$, so the bookkeeping remainder H_{dual} cancels the local curvature to better than 99.99% (Observation 5.1). Direct evaluation at the rounded ordinate $t = 14.13$, at distance approximately 5×10^{-3} from the first non-trivial zero $\gamma_1 \approx 14.1347$, gives $H_\xi(\frac{1}{2}, t) \approx +89\,577.92$, illustrating the singular kernel $(s - \rho)^{-2}$ that dominates near zeros (Observation 5.2).

What remains open. Whether the singular test function implicit in H_ξ — whose Mellin side produces the kernel $(s - \rho)^{-2}$ — can be embedded into, or approximated by, the class of admissible test functions for which Weil positivity is equivalent to the Riemann Hypothesis (Open Problem 8.1).

Structure. The formal core in §§ 2–4 is non-circular and proof-complete. No use of the Riemann Hypothesis, the GUE conjecture, or the Hilbert–Pólya postulate is made anywhere in §§ 2–4.

Notation and Conventions

Critical line. Throughout, $s = \sigma + it$ with $\sigma = \Re(s)$ and $t = \Im(s)$; the critical line means $\sigma = \frac{1}{2}$. The completed zeta function is $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$, so that $\xi(s) = \xi(1-s)$ on the whole plane.

Global parameters. The variables of this paper are $\sigma = \Re(s) > 0$, $t \in \mathbb{R}$ (a real imaginary part), and the prime cutoff $\kappa \geq 2$ (varied across the divergence and convergence statements). The cutoff κ runs over integers when summing $\sum_{p \leq \kappa}$ with p prime; equivalently, κ indexes the prime $p_{\pi(\kappa)}$. No further parameters enter the formal core.

Main objects. The curvature field¹ is

$$H_\xi(\sigma, t) := \partial_\sigma^2 \log |\xi(\sigma + it)|^2,$$

defined for $s = \sigma + it$ away from the zeros of ξ (at a zero, the kernel $(s - \rho)^{-2}$ is singular; cf. Observation 5.2). The truncated local curvature at cutoff κ is

$$H_{\text{local}}(\sigma, \kappa) := \psi_1(\sigma) + \sum_{p \leq \kappa} V_p(\sigma),$$

with archimedean contribution $\psi_1(\sigma) = \frac{1}{8}\psi^{(1)}(\sigma/2)$ ($\psi^{(1)}$ the trigamma function) and local Euler-factor contribution $V_p(\sigma) = 4(\log p)^2 p^{-2\sigma} / (1 - p^{-2\sigma})^2$. Here “archimedean contribution” refers to the Gamma-factor contribution in the local curvature model; the polynomial factor $s(s-1)$ in the completed function $\xi(s)$ is not included in H_{local} . The bookkeeping remainder is $H_{\text{dual}}(\sigma, t, \kappa) := H_\xi(\sigma, t) - H_{\text{local}}(\sigma, \kappa)$. The non-trivial zeros of $\zeta(s)$ are denoted ρ ; the first few imaginary ordinates are $\gamma_1 \approx 14.135$, $\gamma_2 \approx 21.022$.

Analytical foundations. The formal core uses the Hadamard product representation of ξ , log-convexity of the Gamma function (Bohr–Mollerup; Artin [9]), the prime number theorem via a Mertens-type partial-summation estimate (Montgomery–Vaughan [10]), and the Tate local Mellin-norm framework [15] for the Euler-factor contributions.

Numerical computations. All numerical results were computed with `mpmath` at 50 decimal digits of working precision; Riemann zeros γ_k were evaluated via `mpmath.zetazero`. Verification code is available at <https://github.com/utehrani/analysislab-nt>.

Not to be confused with. The truncated local curvature $H_{\text{local}}(\sigma, \kappa)$ used in this paper is the prime-side sum of squared local Euler-factor contributions associated with the second-derivative specialisation of the explicit formula; it is not the classical von Mangoldt prime sum $\sum_{p^k \leq x} \Lambda(p^k)$ and not the renormalised prime weight that appears in admissible-test-function constructions for the Weil functional. The local positivity $V_p(\sigma) \geq 0$ established below is a place-by-place positivity statement and does *not* imply Li positivity of the coefficients λ_n (§ 7).

¹We use “curvature” as shorthand for $\partial_\sigma^2 \log |\xi|^2$ without any Riemannian, Kähler, or Arakelov content. This should be distinguished from the Frenet planar curvature of $t \mapsto \zeta(\sigma + it)$ studied by Sourmelidis–Steuding [24] and from the Quillen metric second variation of Selberg zeta functions (Fedosova–Rowlett–Zhang [25]).

1 Introduction

1.1 Background and Motivation

This is the first paper in a series developing an explicit, finite-cutoff framework for studying the distribution of the non-trivial zeros of the Riemann zeta function $\zeta(s)$. The organising idea of the series is that prime-side quantities at finite cutoff carry arithmetic information about the zero geometry that is accessible by classical analytic means. The present paper introduces the central object of the series, the prime-cutoff decomposition of the second logarithmic derivative of the completed zeta function ξ .

The Hadamard product representation of the entire function ξ , together with logarithmic differentiation, gives

$$(\log \xi)''(s) = - \sum_{\rho} \frac{1}{(s - \rho)^2}, \quad (1)$$

where the sum runs over the non-trivial zeros ρ of ζ . The real part of (1), evaluated as a function of σ at fixed t , defines the curvature field H_{ξ} . Stopple [17] studies the second logarithmic derivative of $\zeta(s)$ as a meromorphic function and analyses the location of its complex zeros. Our paper restricts to the real variable σ and decomposes $H_{\xi}(\sigma, t)$ via the explicit formula; the two studies are complementary. On the local (prime and archimedean) side, the present paper introduces a separately constructed quantity, the truncated local curvature $H_{\text{local}}(\sigma, \kappa)$, built in the Tate framework from per-place log-Mellin-norm contributions: an archimedean trigamma term $\psi_1(\sigma)$ and a finite Euler-factor sum $\sum_{p \leq \kappa} V_p(\sigma)$ with closed forms specified in §2. The bookkeeping remainder is defined by $H_{\text{dual}}(\sigma, t, \kappa) = H_{\xi}(\sigma, t) - H_{\text{local}}(\sigma, \kappa)$; in this paper it serves only as a bookkeeping device for the decomposition, and no spectral interpretation is imposed on it.

The identity $H_{\xi} = H_{\text{local}} + H_{\text{dual}}$ holds by definition of H_{dual} ; its interpretation as a curvature-form explicit-formula reorganisation is formal at the level of singular test functions, specialised to the kernel $2\Re(s - \rho)^{-2}$. The classical reference for the explicit formula and its specialisations is Edwards [11]. Burnol’s invariant-analysis approach to the explicit formula [7] treats the local terms uniformly across all places, and the Connes–Consani trace-formula work at the archimedean place [2] establishes positivity statements there for compactly supported test functions. Neither of these isolates the prime-cutoff second-derivative decomposition $H_{\xi} = H_{\text{local}} + H_{\text{dual}}$ used here, in which, at every finite cutoff κ , the truncated local curvature $H_{\text{local}}(\sigma, \kappa)$ is the sum of the archimedean trigamma term $\psi_1(\sigma)$ and the explicit closed-form Euler-factor contributions $V_p(\sigma)$. The principle of decomposing a global zeta-related object as a sum over places via local Mellin/Tate analysis is extensively developed, from Tate’s thesis [15] through Haran [21] and Burnol’s conductor-operator papers [7] to Connes–Consani’s semi-local framework [2]. Our paper decomposes the real-variable function $H_{\xi}(\sigma, t)$ with finite prime cutoff, a two-parameter picture not present in those works. Equivalently, the decomposition $H_{\xi} = H_{\text{local}} + H_{\text{dual}}$ is the classical explicit formula [11] truncated to primes $p \leq \kappa$ and applied *formally* — at the level of singular test functions and distributions — to the test function whose Mellin side produces the kernel $(s - \rho)^{-2}$. Rigorous embedding of this singular

test function into the admissible class S_{ad} of the Weil functional is the subject of Open Problem 8.1. At the level of the formal identity, H_{local} is the truncated local side, while H_{dual} is the residual term required to balance the zero-side curvature field after subtracting this truncated local contribution.

The present construction, whose finite-place positivity is the main result of this note, should be distinguished from established operator-theoretic programmes for the Riemann Hypothesis: it is not a Hilbert–Pólya spectral realisation in the sense of Connes [1], nor a Berry–Keating type quantum-Hamiltonian construction [12], nor a de Branges structure-function model [13, 14]. The vocabulary of “curvature” used here refers solely to the second-derivative origin of $V_p(\sigma)$ in $\log|\xi(\sigma+it)|^2$; no further geometric content is asserted. To the best of our knowledge, the prime-cutoff curvature formulation $H_\xi = H_{\text{local}} + H_{\text{dual}}$ does not appear explicitly in the literature (cf. J.-F. Burnol, personal communication).

1.2 Main Results

The four formal statements of this paper concern the prime-side object $H_{\text{local}}(\sigma, \kappa)$ alone. First, $V_p(\sigma) \geq 0$ at every prime p and every $\sigma > 0$ (§3, Observation 3.1), hence $H_{\text{local}}(\sigma, \kappa) \geq 0$ throughout (Corollary 3.2). Second, on the critical line, the truncated local curvature diverges with order $(\log \kappa)^2$ as $\kappa \rightarrow \infty$ (§4, Lemma 4.1). Third, above the critical line the truncated local curvature tends to a finite limit (Lemma 4.2). The critical line is therefore the threshold separating divergence from convergence of the local curvature sum.

The numerical illustrations of §5 quantify the cancellation between H_{local} and H_{dual} off the critical line, and the singular peak of H_ξ near non-trivial zeros.

The Rankin–Selberg discussion of §6 and the Li-coefficient comparison of §7 are informational only. Their role is to place the curvature decomposition within the classical analytic-number-theory landscape and to delimit explicitly what local positivity does *not* imply.

The single open problem (§8, Open Problem 8.1) asks whether the singular test function implicit in H_ξ can be embedded into the class of admissible test functions for which Weil positivity is equivalent to the Riemann Hypothesis.

1.3 Reader Contract

This paper *proves* four statements about the truncated local curvature: place-by-place positivity at finite primes (Observation 3.1); non-negativity of the truncated sum $H_{\text{local}}(\sigma, \kappa)$ (Corollary 3.2); critical divergence of order $(\log \kappa)^2$ on $\sigma = \frac{1}{2}$ (Lemma 4.1); and convergence to a finite limit for $\sigma > \frac{1}{2}$ (Lemma 4.2). These are unconditional mathematical statements.

This paper *establishes numerically* two finite computations from the stated parameters (Observations 5.1–5.2): the off-critical compensation between H_{local} and H_{dual} at $(\sigma, t, \kappa) = (0.3, 0, 1300)$, and the singular peak of H_ξ near γ_1 on the critical line. Both numerical statements are finite computations from the stated parameters and are labelled [numerical] in their theorem headings.

This paper *leaves open* the embedding of the singular test function implicit in H_ξ into the admissible test-function class of the Weil functional (Open Problem 8.1, § 8). A positive answer would connect the curvature decomposition to the Weil positivity framework; whether such a connection extends to a full Weil positivity statement on the relevant admissible class is a separate and deeper question.

2 The Curvature Field

The Hadamard product representation of the entire function $\xi(s)$, together with logarithmic differentiation, yields

$$H_\xi(\sigma, t) := \frac{d^2}{d\sigma^2} \log|\xi(\sigma + it)|^2 = 2 \Re(\log \xi)''(s) = -2 \Re \sum_{\rho} \frac{1}{(s - \rho)^2}, \quad (2)$$

where the sum runs over the non-trivial zeros ρ of $\zeta(s)$ and $s = \sigma + it$. The identity $\partial_\sigma^2 \log|\xi(\sigma + it)|^2 = 2 \Re[(\xi'/\xi)'(\sigma + it)]$ is explicit in Goldštejn–Grigutis [16], who study the sign of $\Re(\xi'/\xi)$ on horizontal lines; our paper uses the same identity to decompose the σ -Hessian into local and dual parts. The quantity H_ξ is the spectral (zero) side of an explicit-formula identity at second-derivative level: formally choosing a test function — at the level of singular or distributional test functions — whose Mellin transform produces the kernel $(s - \rho)^{-2}$ recovers H_ξ as the spectral output (rigorous treatment: Open Problem 8.1).

The local (prime and archimedean) side is constructed separately, in the adelic framework of Tate’s thesis, as a sum of per-place log-Mellin-norm contributions. At the archimedean place,

$$\psi_1(\sigma) := \frac{1}{4} \frac{d^2}{d\sigma^2} \log|\Gamma(\sigma/2)|^2. \quad (3)$$

For real σ this coincides with $\frac{1}{8}\psi^{(1)}(\sigma/2)$, where $\psi^{(1)}$ denotes the trigamma function; one has $\frac{d^2}{d\sigma^2} \log|\Gamma(\sigma/2)|^2 = \frac{1}{2}\psi^{(1)}(\sigma/2)$ by the chain rule, so $\psi_1(\sigma) = \frac{1}{4} \cdot \frac{1}{2}\psi^{(1)}(\sigma/2) = \frac{1}{8}\psi^{(1)}(\sigma/2)$. At each finite place p , with M_s the local Mellin transform and f_p the standard Tate test function,

$$V_p(\sigma) := \frac{d^2}{d\sigma^2} \log\|M_s f_p\|_{L^2(\mathbb{Q}_p^\times)}^2. \quad (4)$$

Explicitly, for the unramified local factor, $\|M_s f_p\|^2 = (1 - p^{-1})(1 - p^{-2\sigma})^{-1}$, so that

$$V_p(\sigma) = 4(\log p)^2 \frac{p^{-2\sigma}}{(1 - p^{-2\sigma})^2}. \quad (5)$$

The truncated local curvature at cutoff κ is

$$H_{\text{local}}(\sigma, \kappa) := \psi_1(\sigma) + \sum_{p \leq \kappa} V_p(\sigma), \quad (6)$$

and the bookkeeping remainder is

$$H_{\text{dual}}(\sigma, t, \kappa) := H_{\xi}(\sigma, t) - H_{\text{local}}(\sigma, \kappa). \quad (7)$$

For $\sigma > \frac{1}{2}$, $H_{\text{local}}(\sigma, \kappa)$ converges as $\kappa \rightarrow \infty$ (Lemma 4.2) and $H_{\text{dual}}(\sigma, t, \kappa)$ has a well-defined limit in κ . At $\sigma = \frac{1}{2}$, H_{local} diverges (Lemma 4.1); the decomposition $H_{\xi} = H_{\text{local}} + H_{\text{dual}}$ is then interpreted at every finite κ , and H_{dual} is treated as a bookkeeping object without spectral interpretation.

The decomposition $H_{\xi} = H_{\text{local}} + H_{\text{dual}}$ holds at every finite cutoff κ by the definition of H_{dual} in (7). Read as a formal identity, it is the second-derivative specialisation of the truncated explicit formula to the singular test function whose Mellin side produces $(s - \rho)^{-2}$; rigorous treatment of that test function is the subject of Open Problem 8.1. The identity holds by definition at every finite cutoff κ ; the interpretation as a curvature-form explicit-formula reorganisation is formal.

3 Local Positivity

Observation 3.1 (Local positivity at finite places; proved). *For every prime p and every $\sigma > 0$,*

$$V_p(\sigma) \geq 0.$$

Proof. From the explicit expression (5),

$$V_p(\sigma) = 4(\log p)^2 \frac{p^{-2\sigma}}{(1 - p^{-2\sigma})^2} = \left(\frac{2(\log p) p^{-\sigma}}{1 - p^{-2\sigma}} \right)^2 \geq 0.$$

This is immediate: $V_p(\sigma)$ is a perfect square for every $\sigma > 0$, a special case of the classical log-convexity of Dirichlet series with non-negative coefficients. The content of the observation is the identification of V_p as the σ -Hessian of $\log \|M_s f_p\|^2$ in Tate's local framework [15]. $\square$

The archimedean contribution $\psi_1(\sigma) \geq 0$ is classical, following from the log-convexity of the Gamma function (Bohr–Møllerup theorem; cf. Artin [9]). Combining the two place-wise positivities yields the following.

Corollary 3.2 (Truncated local curvature is non-negative; proved). *For every $\sigma > 0$ and every $\kappa \geq 2$,*

$$H_{\text{local}}(\sigma, \kappa) \geq 0.$$

Place-by-place positivity does not by itself approach the Riemann Hypothesis (cf. § 7); its role in this paper is to make the prime-side mass of H_{ξ} explicit, so that the cancellation between H_{local} and the bookkeeping remainder H_{dual} becomes visible at every finite cutoff. The numerical observations of § 5 quantify this cancellation off the critical line, where it is near-total, and on the critical line, where H_{ξ} exhibits singular peaks near the non-trivial zeros.

4 Divergence and Convergence

The behaviour of the truncated local curvature as $\kappa \rightarrow \infty$ separates the half-plane into two regimes. On the critical line, H_{local} diverges with order $(\log \kappa)^2$; for every $\sigma > \frac{1}{2}$, H_{local} tends to a finite limit.

Lemma 4.1 (Critical divergence; proved). *As $\kappa \rightarrow \infty$,*

$$H_{\text{local}}(\tfrac{1}{2}, \kappa) \asymp (\log \kappa)^2.$$

Proof. From (5), $V_p(\frac{1}{2}) = 4(\log p)^2 p^{-1}(1 - p^{-1})^{-2}$, so for large p ,

$$V_p(\tfrac{1}{2}) = 4(\log p)^2 p^{-1} (1 + o(1)).$$

Partial summation against the prime number theorem $\pi(x) \sim x/\log x$ yields the standard Mertens-type estimate

$$\sum_{p \leq \kappa} \frac{(\log p)^2}{p} = \tfrac{1}{2}(\log \kappa)^2 + O(\log \kappa) \quad (\kappa \rightarrow \infty)$$

(Montgomery–Vaughan [10], § 2.2), hence $\sum_{p \leq \kappa} V_p(\frac{1}{2}) \asymp (\log \kappa)^2$. The archimedean contribution $\psi_1(\frac{1}{2})$ is bounded, so $H_{\text{local}}(\frac{1}{2}, \kappa) \asymp (\log \kappa)^2$. Moreover,

$$V_p(\tfrac{1}{2}) = 4 \frac{(\log p)^2}{p} + O\left(\frac{(\log p)^2}{p^2}\right),$$

and the error term is summable over primes. Hence the same Mertens-type estimate gives the sharper asymptotic

$$H_{\text{local}}(\tfrac{1}{2}, \kappa) = 2(\log \kappa)^2 + O(\log \kappa),$$

with the bounded archimedean term absorbed into the error. In particular, $H_{\text{local}}(\frac{1}{2}, \kappa) \asymp (\log \kappa)^2$. $\square$

Lemma 4.2 (Subcritical convergence; proved). *For every $\sigma > \frac{1}{2}$,*

$$H_{\text{local}}(\sigma, \kappa) \rightarrow H_{\text{local}}(\sigma, \infty) < \infty \quad (\kappa \rightarrow \infty).$$

Proof. From (5), $V_p(\sigma) = O((\log p)^2 p^{-2\sigma})$. For $\sigma > \frac{1}{2}$ one has $2\sigma > 1$, hence the series $\sum_p (\log p)^2 p^{-2\sigma}$ converges absolutely, by comparison with $\sum_p p^{-2\sigma+\delta}$ for any $\delta \in (0, 2\sigma - 1)$. The archimedean contribution $\psi_1(\sigma)$ is finite for $\sigma > 0$. The result follows. $\square$

The critical line is therefore the threshold separating divergence from convergence of the local curvature sum. For completeness, the same comparison, again by the prime number theorem, gives divergence for $0 < \sigma < \frac{1}{2}$, since then $\sum_p (\log p)^2 p^{-2\sigma}$ diverges. Thus $\sigma = \frac{1}{2}$ is the boundary between convergence to the right and divergence on and to the left of the critical line.

Matiyasevich, Saidak, and Zvengrowski [20] identify $\sigma = \frac{1}{2}$ as the conjectural boundary of

horizontal monotonicity of $|\zeta(\sigma + it)|$ via the first σ -derivative (see also Sondow–Dumitrescu [26] for a related monotonicity reformulation of RH). Lemmas 4.1 and 4.2 establish the same critical line as the divergence/convergence threshold of $H_{\text{local}}(\sigma, \kappa)$ via the second σ -derivative; the two results are complementary.

5 Numerical Illustration

The two observations of this section quantify the cancellation between H_{local} and H_{dual} at a representative off-critical point and the singular peak of H_{ξ} near a non-trivial zero on the critical line.

Observation 5.1 (Off-critical compensation; numerical). *At $(\sigma, t) = (0.3, 0)$ with $\kappa = 1300$,*

$$H_{\text{local}} \approx 915, \quad H_{\xi} \approx 0.09.$$

The bookkeeping remainder H_{dual} therefore cancels H_{local} to better than 99.99%.

Observation 5.2 (Sampled values near zeros; numerical). *Take the rounded ordinates $t_1 = 14.13$, at distance approximately 5×10^{-3} from the first non-trivial zero $\gamma_1 \approx 14.1347$, and $t_2 = 21.02$, at distance approximately 2×10^{-3} from the second non-trivial zero $\gamma_2 \approx 21.0220$. Direct evaluation of $H_{\xi}(\sigma, t) := \partial_{\sigma}^2 \log|\xi(\sigma + it)|^2$ at these ordinates gives*

$$H_{\xi}(0.3, t_1) \approx -49.78, \quad H_{\xi}(\tfrac{1}{2}, t_1) \approx +89\,577.92,$$

and on the critical line near the second zero, $H_{\xi}(\tfrac{1}{2}, t_2) \approx +480\,754.91$.

The spike at $\sigma = \frac{1}{2}$ arises because the kernel $2\Re(s - \rho)^{-2}$ in (2) has a double pole at $s = \rho$; near a zero this term dominates the spectral side, producing the large values observed. The larger value at t_2 in Observation 5.2 reflects the smaller sampling distance $|t_2 - \gamma_2| \approx 2 \times 10^{-3}$ to γ_2 , compared with $|t_1 - \gamma_1| \approx 5 \times 10^{-3}$ to γ_1 , together with the $|s - \rho|^{-2}$ singularity of the kernel. The functional-equation symmetry $H_{\xi}(\sigma, t) = H_{\xi}(1 - \sigma, t)$, which follows from $\xi(s) = \xi(1 - s)$, holds to machine precision in all numerical evaluations performed.

Together, Observations 5.1 and 5.2 quantify the cancellation between the place-by-place non-negative H_{local} (Observation 3.1, Corollary 3.2) and the bookkeeping remainder H_{dual} : numerically, the cancellation reaches better than 99.99% off the critical line, while on the critical line H_{ξ} takes large positive values when (σ, t) is sampled close to a non-trivial zero, consistent with the singular kernel $(s - \rho)^{-2}$. The relationship is observational at finite cutoff and is not asserted as a theorem.

6 Rankin–Selberg Analogue

The divergence at $\sigma = \frac{1}{2}$ established in Lemma 4.1 is not specific to $\zeta(s)$. This section is informational. Under the standard Rankin–Selberg prime-number/Tauberian framework for

a cuspidal automorphic representation π , and after removing ramified primes and separating prime-power terms, one expects the weighted prime sum

$$\sum_{p \leq \kappa} |\lambda_\pi(p)|^2 \frac{(\log p)^2}{p}$$

to have order $(\log \kappa)^2$, with a positive leading constant depending on the normalisation of the Rankin–Selberg residue (cf. Kowalski–Michel [8] for related prime sums in the Rankin–Selberg context). This is not used in any proof of the present paper.

The Rankin–Selberg L -function extension of Li’s criterion has been studied by Smajlović [22] and Odžak–Smajlović [23]. The analogy in the present paper is formal and motivational; no Li-criterion proof is claimed here.

7 Connection to Li Coefficients

The Li coefficients λ_n , defined via

$$g(z) := \log \xi \left(\frac{1}{1-z} \right), \quad g'(z) = \sum_{n \geq 1} \lambda_n z^{n-1},$$

satisfy Li’s criterion [6]:

$$\text{RH} \iff \lambda_n \geq 0 \quad \text{for all } n \geq 1.$$

Differentiating once more,

$$g''(z) = (\log \xi)'' \left(\frac{1}{1-z} \right) (1-z)^{-4} + 2(\log \xi)' \left(\frac{1}{1-z} \right) (1-z)^{-3}, \quad (8)$$

so that the term $(\log \xi)''(s)$ appearing in (8) is exactly the core of the curvature field $H_\xi(s) = 2\Re(\log \xi)''(s)$. The present curvature quantities are therefore structurally close to Li-type positivity criteria: both are built from logarithmic derivatives of ξ . The difference is that H_ξ is a continuous pointwise second-derivative field, whereas the Li coefficients package the same logarithmic data into a discrete sequence after the Möbius reparametrisation around $s = 1$.

Remark. *The local positivity $V_p(\sigma) \geq 0$ at each finite place established in Observation 3.1 does not imply $\lambda_n \geq 0$. Li positivity is a global statement mixing all places, the functional equation, the complete zero geometry, and a specific test-function structure on the disc parametrisation. Place-by-place positivity is a compatible local input but not sufficient for Li positivity, and the gap between the two is of Riemann-Hypothesis strength.*

8 Open Problems

The central open problem arising from the curvature decomposition of this paper is the following.

Open Problem 8.1 (Embedding into the Weil framework). *Does there exist a global packaging — a summation procedure, test-function class, or regularisation — that transfers the local positivity*

$$V_p(\sigma) \geq 0 \ (\forall p \text{ prime}), \quad \psi_1(\sigma) \geq 0,$$

together with the critical divergence $H_{\text{local}}(\frac{1}{2}, \kappa) \rightarrow \infty$, into a global Li- or Weil-type positivity statement?

*More precisely: can the singular test function implicit in H_ξ — whose Mellin side produces the kernel $(s - \rho)^{-2}$ — be embedded in, or approximated by, the class S_{ad} of admissible test functions for which Weil positivity $W(f * \check{f}) \geq 0$ is equivalent to the Riemann Hypothesis?*

We note that Weil positivity is highly sensitive to the choice of test-function class and regularisation. Bombieri [18] studies Weil’s quadratic functional on support-restricted L^2 -spaces and its finite-dimensional truncations, while Lagarias [19] formulates the Li/Weil framework on test spaces with explicit analyticity and decay restrictions. These results show that one cannot pass from a formal explicit-formula kernel to a positivity statement without controlling the admissible class and the limiting process. The kernel implicit in the present paper, whose Mellin side produces $(s - \rho)^{-2}$, is more singular than the first-order distributions appearing in the classical explicit formula. The problem of embedding or approximating it inside an admissible Weil framework is therefore a genuine regularisation problem, not a formal consequence of local positivity.

A positive answer to Open Problem 8.1 would connect the curvature decomposition to the Weil positivity framework (Weil [3]; Lagarias [4]; Connes–Consani [2]). By Weil’s criterion (Weil [3]; Bombieri–Lagarias [5]), positivity of the Weil quadratic functional on the full relevant admissible class is equivalent to the Riemann Hypothesis. Positivity for the particular family arising from the present construction would therefore be only a family-level positivity statement, not by itself an RH criterion. The present Open Problem is therefore the prior step of fitting $H_{\text{local}}(\sigma, \kappa)$ into the admissible framework, and is not the positivity step itself; the latter remains a separate and deeper open question, and is not addressed here.

Non-Circularity

We document explicitly that the results of this paper do not rely on the objects they are directed toward.

No RH as input. The Riemann Hypothesis is not assumed at any point. The non-trivial zeros ρ enter the curvature field H_ξ through the Hadamard product representation of ξ , which is unconditional; no property of the location of the zeros is used in the formal core §§ 2–4.

No GUE, Montgomery, or Hilbert–Pólya hypothesis. No correlation statistic of the zeros and no postulated spectral realisation of the zeros is assumed or used. The terminology of “curvature” refers solely to the second-derivative origin of $V_p(\sigma)$ in $\log|\xi(\sigma + it)|^2$ and is not used as a geometric or operator-theoretic input.

No Li positivity as input. The Li criterion of § 7 is cited only as a comparison object; positivity of the coefficients λ_n is not assumed and not used in any proof of this paper. The Remark in § 7 explicitly states that local positivity does not imply Li positivity.

No Weil positivity as input. The Weil functional $W(f * \check{f})$ and the admissible class S_{ad} enter only in the formulation of Open Problem 8.1. No positivity statement on W is assumed in any of Observation 3.1, Lemmas 4.1, 4.2, or in Corollary 3.2.

Inputs from classical analytic number theory. The formal core of the paper uses only standard inputs: the Hadamard product for ξ , log-convexity of the Gamma function (Bohr–Mollerup), the prime number theorem $\pi(x) \sim x/\log x$, and elementary partial summation. The Rankin–Selberg discussion in § 6 relies additionally on the simple pole of $L(s, \pi \times \tilde{\pi})$ at $s = 1$ and a Tauberian argument; this section is informational and not used in the formal core.

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Call for Feedback

This paper is part of an open research initiative on prime–zero coupling and explicit-formula operators. All papers in the series are freely available on Zenodo under CC BY 4.0, and the accompanying verification code is hosted on GitHub.

Zenodo:

https://zenodo.org/search?q=metadata.creators.person_or_org.name%3A%22Tehrani%2C%20Ulrich%22

GitHub:

<https://github.com/utehrani/analysislab-nt>

Topics of particular interest for diagnostic feedback on this note include: (i) whether the singular kernel $(s - \rho)^{-2}$ underlying H_ξ admits a natural regularisation or limit construction in S_{ad} , in the sense of Open Problem 8.1; and (ii) whether the Rankin–Selberg analogue (§ 6) admits a refined asymptotic with explicit leading constant under standard conventions.

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